Theodor Hellqvist, Thesis report

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Availability through technology – A study on how AI could be used Within ILS to increase availability of ground vehicles.

Abstract

This thesis explores how Artificial Intelligence and machine learning can create military utility within Integrated Logistics Support processes to increase the availability of ground vehicles in the Swedish Armed Forces. As modern military logistics face demands for higher efficiency and resilience, AI offers potential solutions, but also introduces socio-technical risks. Through a qualitative, descriptive case study based on seven expert interviews, this study analyzes the phenomenon using the Military Utility framework (Effectiveness, Suitability, Affordability).

The findings indicate that AI has high potential effectiveness. Reducing Mean Time to Repair via Large Language Models used for searching technical manuals, and enabling predictive maintenance through sensor analysis. Affordability is enhanced by AI's ability to run robust simulations for material demand, challenging vulnerable "just-in-time" supply chains. However, Military Suitability is currently constrained by poor data standardization for ground vehicles, strict information security demands, and organizational skepticism toward "black box" technology. Furthermore, the study highlights critical risks, such as "peacetime bias" in training data and "automation bias", where personnel risk losing fundamental mechanical competence. Conclusively, to safely leverage its military utility, AI must be implemented strictly as an advisory decision-support tool, keeping the human in the loop.

Keywords: Integrated Logistics Support (ILS), Artificial Intelligence (AI), Machine Learning, Predictive Maintenance, Ground Vehicles, Military Utility, Automation Bias.

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Tillgänglighet genom teknologi – En studie om hur AI kan användas inom ILS för att öka tillgängligheten av markfordon.

Sammanfattning

Denna uppsats undersöker hur artificiell intelligens och maskininlärning kan skapa militär nytta inom processerna för integrerat logistikstöd för att öka tillgängligheten hos markfordon inom Försvarsmakten. Eftersom modern militär logistik ställs inför krav på ökad effektivitet och motståndskraft erbjuder AI potentiella lösningar, men medför samtidigt socio-tekniska risker. Genom en kvalitativ, beskrivande fallstudie baserad på sju expertintervjuer analyserar denna studie fenomenet med hjälp av ramverket för militär nytta (effektivitet, lämplighet, prisvärdhet).

Resultaten indikerar att AI har hög potentiell effektivitet. Genom att minska den genomsnittliga reparationstiden med hjälp av stora språkmodeller som används för att söka i tekniska manualer, samt möjliggöra förebyggande underhåll genom sensoranalys. Prisvärdheten förbättras genom AI:s förmåga att köra robusta simuleringar för materielbehov, vilket utmanar sårbara ”just-in-time”-försörjningskedjor. Den militära lämpligheten begränsas dock för närvarande av bristfällig standardisering av data för markfordon, strikta krav på informationssäkerhet och organisatorisk skepsis mot ”black box”-teknik. Dessutom belyser studien kritiska risker, såsom ”fredstidsbias” i träningsdata och ”automatiseringsbias”, där personalen riskerar att förlora grundläggande kompetens. Sammanfattningsvis måste AI, för att säkert kunna utnyttja sin militära nytta, implementeras strikt som ett rådgivande beslutsstödverktyg, där människan hålls kvar i beslutsprocessen.

Nyckelord: Integrerat logistikstöd (ILS), artificiell intelligens (AI), maskininlärning, förebyggande underhåll, markfordon, Militär Nytta, automatisering.

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Glossary	
AI	Artificial intelligence.
ML	Machine Learning
ILS	Integrated Logistics Support
EOI	Element Of Interest
PDM	Predictive maintenance
LLM	Large Language Model
MTTR	Mean Time to Repair
MTBF	Mean Time Between Failures
DOTMPLFI	Doctrine, Organization, Training, Materiel, Personnel, Leadership, Facilities, and Interoperability
SwAF	Swedish Armed Forces
FMV	Swedish Defense Materiel Administration (Försvarets materielverk)
UGV	Unmanned Ground Vehicle

1 Introduction

This chapter introduces the concepts of AI and integrated logistics support. Definitions for both are presented and discussed. Why AI is interesting from a military logistics perspective is also introduced and discussed.

Several research questions are presented, and lastly how the thesis is delimited and what research gap the thesis fills is explained.

1.1 Background

The European Commission's *ReArm Europe* Plan aims to leverage over €800 billion in defense spending, reflecting an unprecedented commitment to domestic defense capabilities (Clapp et al., 2025, p. 3). Sweden is no exception to this, rearming and investing heavily in its own defense in line with the war in Ukraine (Törnmalm & Williams, 2025). The Ukrainian armed forces are often viewed as innovative in their ways of using new technology (Shumlianskyi et al. 2025, p. 2), using drone swarms powered by Artificial Intelligence [AI], which have proven to be very effective against conventional warfighting methods (Askelin, 2025b). AI plays a key part in the war (Myronyshena 2026), and since the release of the large language model [LLM] ChatGPT in 2022, AI has seen a massive boost in public interest. Students use it to write essays, engineers use it to write code (Chatterji et al., 2025, pp. 1-2), and drones use it to fly.

While data collection and analysis of the battlefield have in turn improved due to defense systems becoming increasingly digitalized (Kufakunesu et al. 2025), the same systems have also become more complex and expensive. The concept of integrated logistics support [ILS] was developed as a solution to these issues. NATO (2012, pp.184-185) defines ILS as

> the deliberate integration of systems/equipment logistic support considerations into the system's life cycle management during the outset of the programme/project. ILS prescribes that all elements of logistic support be planned, acquired, tested and provided in a timely and cost-effective manner.

The ILS elements include training of personnel, operations, maintenance, potential redesign and dismantling of systems (NCOSE, 2023, p. 171). ILS provides these systems with robust logistical solutions to be able to support the warfighting effort, which has proved harder to achieve in wartime than expected (Barnett et al, 2023). Some even say that Russia's failures in the war with Ukraine stem from poor logistical planning (Martin Et al. 2023; Bella 2026). Due to this, the US army even named it one of its core "lines". NATO (2025) similarly argues that logistics serves as the backbone of a military's power. This backbone is heavily supported by ground vehicles, transporting soldiers and materiel to and from the front lines (Kinsey & Ti, 2023a). With Sweden's role in NATO being the logistics-hub in the north of Europe (Granlund, 2025), effective logistics will be essential. It can even be argued that working logistics play a bigger factor than a developed strategy in achieving tactical victory (Ti & Kinsey, 2023b).

1.2 Problematization

There is pressure for the SwAF to become more data driven, and to use and rely more on technology, from both themselves and the industry (Mojtahed et al. 2024; Limmergård, 2025). Not just to be able to respond to current and future threats, but to also keep up with other NATO allies. Some even argue that Sweden would have lost a conflict similar to the one in Ukraine with its current processes (Askelin, 2025a, p. 43).

The Swedish Government has therefore highlighted AI as a strategic resource for Sweden's total defense (Regeringskansliet, 2026). There have been several investments in infrastructure (Olin, 2026), and efforts to develop domestic AI capability through AI Sweden (AI Sweden, 2025). Meanwhile, Mojtahed et al. (2024) argues that the SwAF as an organization might not be mature enough to fully implement the latest technologies, due to their reluctance to implement new technology and lack of required technical competence within the organization. Another issue is data quality, as the system is ruled by the data it's trained on (Jonker & Aquino, 2026). If the training data doesn't reflect the real problem or situation at hand, the AI system will make mistakes, which can result in undesired consequences, especially in matters of national security (Yu & Carroll, 2022).

This thesis explores a use case that may not be at the bleeding edge of AI- or technical development, but a military use that has not been thoroughly researched in an academic sense. AI predictive maintenance is most likely already being used by the different air forces to maintain and predict failures of their aircrafts before they happen, and has for some time been used by the US air force (C3 AI, 2025). One can assume the same is being done for the Swedish air force. In regards to land and maritime systems, the same assumption cannot be made. With Sweden having a supporting role, through being the logistics hub in the north of Europe, how can we make sure the supporting logistical systems are available when there is a sudden need? This study will therefore examine how the availability of ground vehicles could be improved using AI.

1.3 Research- Aim and questions

This thesis aims to contribute to the existing research of AI in the military field. The study researches how AI can be used to improve ILS processes using the Military Utility concept as a theoretical framework to determine its value. The thesis uses a deductive methodology, where previous research and theories about AI and ILS are used to formulate the questionnaire for the interviews.

The goal with the thesis is to answer the following research question:

(1) ***How can AI create military utility in ILS processes?***

Sub questions that will be researched are:

(1) What are the risks with becoming more data driven?

1.4 Delimitations

AI used for logistical or maintenance purposes is the main point of study. This essay will not discuss the potential ethical issues with using AI powered weapon systems, but they may be mentioned.

There is no specific AI algorithm or approach that is under investigation. Machine learning AI and Large Language Models are both classified as AI, and will therefore be a topic of discussion. However, different AI or machine learning algorithms and their performance is not compared, rather the potential use cases of AI models are discussed and assessed.

This paper has been researched from a Swedish perspective; other nations perspectives may differ as their starting point, and organizational readiness for AI adoption may be different.

The main defense system under investigation are ground vehicles. While maritime systems may be of interest to study, one is even mentioned, they are not the focus of this thesis. No specific ground vehicle system has been chosen due to time constraints, and to keep it more generalizable. The results may therefore have been different if a specific ground vehicle, or maritime system were the subject of research.

1.5 Research Gap

There have already been some reports and student theses at SEDU, looking into AI and maintenance. This paper builds on that research, but investigates the problem from a broader perspective, looking at the whole integrated logistics support system to try and find use cases.

This paper explores the problem through interviews with armed forces personnel, academia, and from the Swedish defense materiel administration. It’s possible these findings help other NATO countries in navigating AI implementation within their defense logistics. This paper assists in finding and explaining use cases that later can be further researched and investigated. The military field is inherently secretive and difficult to study without insight. This study contributes to discussion and further studies in ways AI can be used to support the longevity of vehicles

2 Previous research

This chapter discusses previous research into the main theme's artificial intelligence and a background into integrated logistics support. The concepts of reliability, availability, and maintainability [RAM] are discussed as well.

2.1 Integrated Logistics Support

Integrated logistics support [ILS] is an approach that enables support of military systems for their entire life cycle, from development to phase out, the procurement of support items, the supply but also the distribution of these items, and the maintenance of the systems throughout their entire lifecycle. The goal with ILS is to increase availability of systems, while decreasing their overall costs (INCOSE, 2023, p. 193). The field of Integrated logistics support emerged in the 1960s in the US, as their military systems started to become too complex and expensive (Babbitt,1975 p.4). The complexity of the systems started to affect the maintenance, and in turn the state of readiness. Initially ILS mainly focused on the maintainability of systems, with four key elements present:

(1) *Military readiness through efficient, cost-effective maintenance;*

(2) *Designed 1n reliability and maintainability;*

(3) *Timely availability of support resources needed for Initial phase of operation;*

(4) *Integrated development of funds requirements for logistics resources.*

These four principles still hold true when developing today's defense systems, and even in civilian systems, such as trains and railways, or aircrafts. The Swedish Defense Materiel Administration (FMV, n.d) explains ILS as" It *involves taking maintenance and operation into account throughout the system's entire lifecycle. The goal is to create systems that have a long service life, require minimal maintenance, and are easy to maintain, when necessary, which also reduces costs over the entire lifecycle*".

Strongly correlated with ILS is reliability, availability, and maintainability, or RAM, which enables the system to perform without failure (INCOSE, 2023, p. 176). These concepts will be addressed below, and how reliability and maintainability affect availability.

2.1.1 Reliability

The same military system can be used in wildly different environments, and they need to be able to perform sufficiently in both extreme heat and extreme cold. A formal definition of reliability could be "*the ability of a system to perform as designed, without failure, in any operational environment, for a stated period of time*" (Tortorella, 2015). Reliability is an engineering field in its own right, for a system to successfully fulfill these demands, the interaction between the environment and its operating conditions need to be deeply understood (INCOSE, 2023, p. 176). There are several ways to measure reliability of products, a common way is to use the Mean Time Between Failures [MTBF] measurement, which looks at how long

the system can be used before failing. The formula is calculated using total operating time divided by the number of failures.

$$MTBF = \frac{Total\ operating\ time}{Number\ of\ failures}$$

While this may be regarded as a key indicator when looking at reliability, it can easily be misused, which turns the reliability engineering effort into a numbers game (INCOSE, 2023, p. 176). The effort is put into achieving a high MTBF in a specific test environment, instead of engineering a system that is inherently reliable.

2.1.2 Maintainability

ILS is closely tied to maintenance, being able to provide workshops with spare parts, or shipping out a critical part to a system on the front lines. A system designed for maintainability tries to reduce all the required maintenance, but also expects the system to be repaired and restored. To be maintainable, the maintenance plan should consider the interchangeability of parts, their accessibility for removal, and their testability. Maintenance can be broken down into four groups (INCOSE, 2023, p. 179).

Corrective maintenance: Unscheduled maintenance performed as a result of failure.

Preventive maintenance: Scheduled maintenance done to retain system performance and availability.

Predictive maintenance: Scheduled maintenance based on system data analysis to estimate and prevent failure.

System upgrades: Performance upgrades to support system life extension.

A common measurement for maintenance is Mean Time to Repair or MTTR. MTTR measures the time to perform a maintenance task or repair.

$$MTTR = \frac{Total\ Time\ Spent\ on\ Repairs}{Number\ of\ Repairs}$$

2.1.3 Availability

The availability of a system is dependent on its maintainability and its reliability (Blanchard, 1998). Availability is the ratio between a system uptime, and the total time of a system, where total time is system uptime plus system downtime. Reliability affects the system uptime, while maintainability affects the system downtime. Downtime is also affected by all the other ILS elements, the number of workshops, their number of staff, their number of spare parts, and their number of suppliers, essentially the whole support environment.

The following formula based on reliability and maintainability can be used to calculate how often a system is available.

$$Availability(\%) = \frac{MTBF}{MTBF + MTTR} \times 100$$

The formula should be taken with a grain of salt, but indicates that if the availability percentage is below a satisfactory number, something about the maintainability or reliability needs to change. It could be that it needs to be maintained more often, or that a part needs to be redesigned to fit a new environment.

Availability can be expressed in a few different ways according to INCOSE (2023, p. 176).

Inherent availability [Ai] is based on the inherent reliability and maintainability of the system, and assumes the system has an optimal support environment.

Achieved availability [Aa] is the same as inherent availability, but the maintenance schedule is taken into account.

Operational availability [Ao] assumes the system is an actual operational environment, and includes logistical and administrative delays.

For this thesis, operational availability is the type of availability used to investigate the phenomenon. The other types of reliability don't take into account how a system is used in a conflict compared to peacetime, where the use or consumption of spare parts may differ in MTBF measurements.

2.2 Artificial intelligence

In recent years, interest and investment in AI has exploded, perhaps due to ChatGPT and other AI chatbots (Sinclair, M. 2026), but also due to its use in military conflicts. Companies like Palantir utilize AI for military targeting support and prioritization.

Artificial intelligence was first coined in the 1950s by John McCarthy. The term machine learning came a few years later, when Arthur Samuel developed a computer program that improved, or learned, the game of checkers over time (Samuel, 1959). AI and Machine learning are often interchangeable terms, and their meaning has changed with the public perception of what AI is. In 2007 AI was defined by McCarthy himself as "*... the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable."* (McCarthy, 2007, p.1)

A more recent definition by Sheihk Et al. (2023), states that AI is" *Systems that display intelligent behavior by analyzing their environment and taking actions – with some degree of autonomy – to achieve specific goals."*

AI can be achieved using several different methods or equations, LLMs or machine learning models. Essentially, it's a system that tries to achieve some goal, autonomously. Investments in AI mainly happen in infrastructure, such as data centers (Munjal & Sachdev, 2025). One company, Allbirds, even went so far as to shift from being a shoe manufacturer, to become an

AI infrastructure provider (Albeck-Ripka, 2026). Which to some may be reminiscent of the 1990s and the early 2000s dot-com bubble, where everything related to ".com" garnered investments and interest (Morris & Alam, 2008). At that time, the technology for the internet came from the defense sector, this time it's the other way around.

AI has been used for some time to reduce costs and improve availability of Aircraft in the civilian and military sector through predictive maintenance (Weisgerber, 2017). Predictive analysis could for example be made through only vibration monitoring, since vibration itself is an indicator of poor health (Molęda et al., 2023). AI technology could also be simply used to do dull, dirty and dangerous tasks, which would free up already scarce personnel to help with more cognitively demanding tasks (Schraagen, 2024).

On the other hand, Bovet (2025, p.139) argues that the growing amounts of data and the current speed at which warfare develops, requires too much of human operators in their decision making. Therefore, support is needed, perhaps in the form of an AI model, that can assist decision-making and reduce cognitive load. Alenljung et al. (2026) explores this use in military planning exercises, the results showed that participants appreciated being able to search through and summarize information, and its ability to generate courses of action.

However, when decisions need to be taken quickly and under pressure, the models may not be so reliable. Wickenborg-Bolin et al., (2026) find in their study that LLMS often show recency bias, deprioritizing the most critical information if it appears first in a list.

2.2.1 AI ILS use in the military

The main issue with implementing AI in the Armed Forces, is the need for expertise in both the military field and the technical field, which is rare and expensive to come by (Ellmén, 2024). Furthermore, trying to become more data driven does not come without risk. For example, in 2017, The US Army Logistics Analysis Center spent $135 million on an AI machine learning model to analyze several sensors and develop tailored maintenance schedules on their Stryker APC fleet (Sayler, 2020 p. 10). However, this may not have had the desired effect. As of 2025, most of the US Army vehicles did not meet the requirements to be classified as mission capable, due to an overall decline in maintenance (Slayton, 2025). Fewer repairs and overhauls – 1,278 overhauls in 2015 compared to just 12 overhauls in 2024 – were performed, and more money had been spent. According to the U.S Government Accountability Office (GAO, 2025, pp. 1-2) this is in part due to supplier shortage, and that for some parts there only exists a single manufacturer, resulting in competition for parts within the US army. Another key issue is the limited number of maintainers and technicians, who had access to technical drawings, and who were well versed in repairs (GAO, 2025, pp. 2-3).

A more recent and perhaps more successful implementation is implementation of predictive maintenance [PdM] for the engine department in five of the Spanish Armada's frigates (Fernández-Barrero et al., 2021). Their PdM system could predict failures before conventional systems notified personnel, but it relies on offline processing of data in distant data centers. Furthermore, Esteban et al. (2026, p.22) investigated how PDM could affect 65 military trucks, also in Spain. That PDM system could predict abnormal behavior up to nine hours before the

conventional system. On the other hand, this solution relied on telemetry data being transmitted through cellular connectivity, which may impact its potential use in combat vehicles.

2.2.2 Trust and adoption of AI

Getting decisionmakers onboard when trying to implement new technology is another issue. Mahale et al. (2024 p.4) argues that explainable AI (XAI) could help in increasing trust and use of the system. Getting the AI model to explain why it does things the way it does through generative AI and natural language. While this may still be some time away, the AI company Antrophic is taking steps towards XAI, by simply using two LLMS simultaneously. Having one model write a response to a prompt, and another model work backwards to interpret and explain the firsts models output (Fraser-Taliente et al. 2026).

Mahale et al. (2024 p.4) also notes that incorporating AI and ML into existing systems could be complex and costly, but at the same time highlights that there are opportunities to develop new cost-effective solutions. While these LLMs might not perform at the same level in general tasks, fine tuning an LLM to fit a specific task, generally yields the same results, with way less computing power (Kamrani et al. 2024).

Different countries have different strategies in place for AI implementation. Some countries have already developed AI doctrines or AI strategies for how their governmental institutions should implement AI (Regeringskansliet, 2026). For example, the US Defense Logistics Agency [DLA], uses AI to predict material demand, optimize inventory, and to monitor supply chain risks (Roberts, 2025).

3 Theoretical framework

This chapter discusses the theoretical framework used in the thesis. To understand what impact AI may have in ILS processes, and what potential risks there are with implementing machine learning in this way, the concept of Military Utility by Andersson et al. (2015) is used. The framework is used to determine what utility an element of interest [EOI] may have in a military setting by looking at three key factors: Effectiveness, Suitability and Affordability, these factors are explained below.

To complement the theory of Military Utility, a socio-technical risk management framework is used. This framework is used to understand the interaction between user, technology, and organization, and some potential risks inherent with implementation of new technology. These two theories are used to try and create a lens which to perceive the EOI with.

3.1 Military Utility

Military Utility is a concept developed by Andersson et al. (2015), which was created to support stakeholders in deciding the use of technology in military activities. This fits well with the goal of this thesis, which is providing decisionmakers some insight into the potential utility of AI within ILS processes in regards to effectiveness, suitability and affordability. The three dimensions are not substitutable, for an EOI to have utility, it has to be effective, suitable and affordable.

For an assessment to be made, knowledge of three variables, the element of interest, the military actor, and a specific context is needed (Andersson et al. 2015, p. 3). In this thesis, the element of interest is AI machine learning, and the context is its use within ILS processes to support vehicle availability. The military actor is the Swedish Defense Materiel Administration [FMV].

Furthermore, Silfverskiöld et al. (2021) argue that, to counteract the limitations with the model, an expert group is needed to bridge the gap. Therefore, actors within the defense sector have been interviewed and their knowledge has been aggregated to support the assessment.

Andersson et al. (2015, p.9) mentions that a technology, such as AI, is more difficult to assess in regards to utility, as it is a more general technology than a ready-made system with AI built in. However, what this thesis aims to examine, is what happens if AI was part of that system, and how that changes the systems potential accessibility.

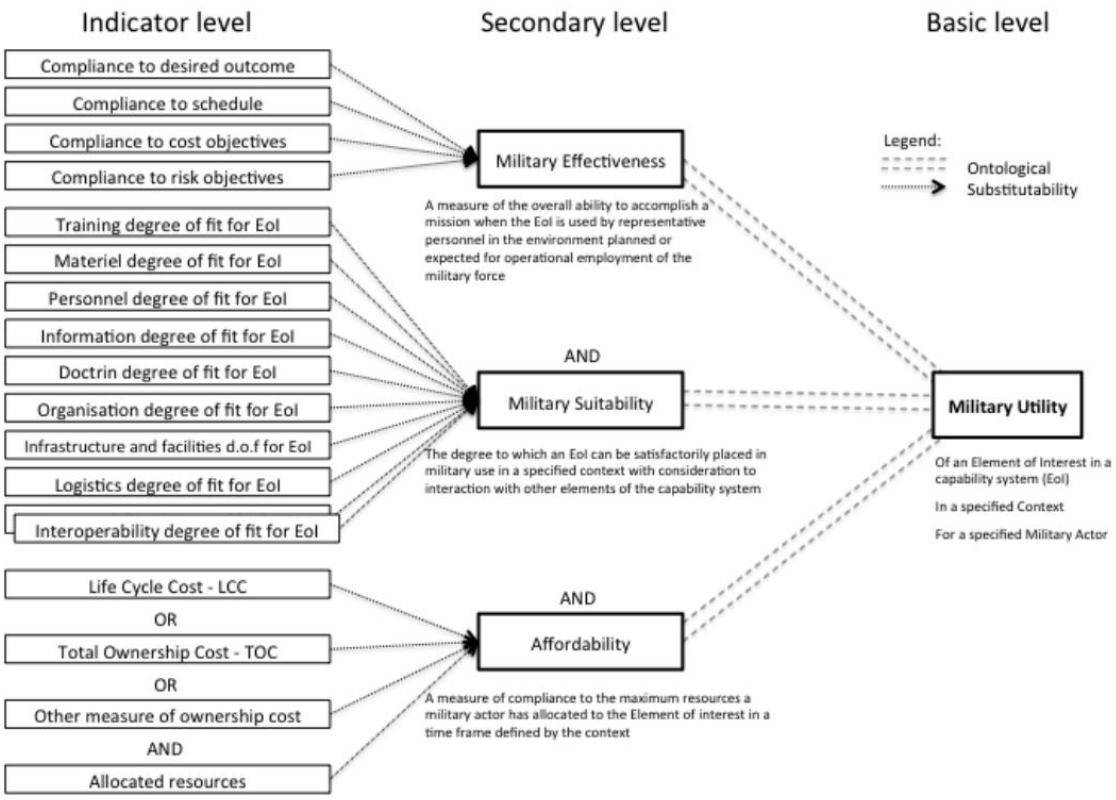


Fig 4. Military Utility concept model (Andersson et al. 2015)

3.1.1 Military effectiveness

Military effectiveness is a measure of the overall ability to accomplish a mission when the Element of Interest (EoI) is used by representative personnel in the environment planned or expected for operational employment. According to Andersson et al. (2015), this dimension is operationalized through four substitutable indicators: compliance to desired outcomes, compliance to schedule, compliance to cost objectives, and compliance to risk objectives.

Since this thesis employs a qualitative approach, effectiveness will not be evaluated through numerical probabilities of mission success. Instead, it will be evaluated through a qualitative analysis of how AI impacts these indicators within ILS processes. For example, "compliance to schedule" can be interpreted as how AI affects time savings and reduces MTTR through enhanced data processing and manual searching. "Compliance to risk" involves the potential hazards of relying on AI, such as the risk of false positives in sensor analysis or the reliance on peacetime data in a wartime context.

3.1.2 Military suitability

Military suitability assesses the degree to which an EoI can be satisfactorily placed into military use within a specified context, considering its interaction with other elements of the capability system. A fundamental degree of suitability is necessary for an EoI to possess any utility; an EoI with low suitability to the other components in the Armed Forces capability system would arguably be of little use.

While Andersson et al. (2015) illustrate this using the TEPIDOIL framework (*Training, Equipment, Personnel, Infrastructure, Concepts and Doctrine, Organization, Information, and Logistics*), this thesis will utilize the DOTMPLFI framework (*Doctrine, Organization, Training, Materiel, Personnel, Leadership, Facilities, and Interoperability*), in alignment with NATO standards, as Sweden now is a part of NATO. In this study, the DOTMPLFI framework acts as a lens to examine potential integration friction.

3.1.3 Affordability

The third dimension, affordability, accounts for the consequences of having limited funding and resources. Andersson et al. (2015) define affordability as a measure of compliance to the maximum resources a military actor has allocated to the EoI in a time frame defined by the context.

Affordability goes beyond mere acquisition costs and is typically operationalized using Life Cycle Cost [LCC] or Total Ownership Cost [TOC]. TOC is particularly relevant as it adds indirect, fixed, and linked costs to the LCC, such as personnel required for administration and supervision. It ensures the solution delivering the best possible desired effect remains affordable to the military actor. In the context of this thesis, affordability will be used to explore how AI might influence both direct life-cycle costs, like supply chain optimization, and hidden alternative costs, such as the value of human labor or the value of human competence.

3.2 Risks

AI systems are inherently socio-technical, Andersson et al. (2023). While risk is sometimes casually conflated with threats and hazards, military and governmental doctrines strictly separate these concepts; a threat is a potential adverse event, whereas a risk is the product of the likelihood of a threat occurring and its subsequent consequences (Försvarsmakten, 2009; Myndigheten för samhällsskydd och beredskap [MSB], 2011)

The military utility framework mentions risks within its “effectiveness” branch. As AI is an overarching technology, risk needs to be evaluated across all socio-technical dimensions, human, technology, and organization, the interactions between these elements represent a breeding ground for risks (Menzefricke et al. 2021). One of the biggest or most common risks with new technology is overreliance on the system's capability, called Automation Bias. Essentially, the human operator relies too heavily on the system and trusts that it always makes the correct decision (Romeo & Conti, 2026). Paradoxically, another significant organizational problem when implementing new technology is getting the employees on board with using and trusting the new system.

Furthermore, the introduction of innovative technologies in military operations often acts as a forcing function that reveals underlying organizational vulnerabilities, sometimes forcing personnel to rely on risky ad-hoc "coping strategies" when these complex systems fail or operate in degraded modes (Johnson, 2012).

4 Method

This chapter presents the way the thesis conducted data collection and what sort of data was collected.

There is also a discussion on the quality of the thesis in regards to validity, reliability and generalizability.

4.1 Research design

For this thesis, a qualitative method was chosen to try and understand how AI machine learning can assist in creating higher vehicle accessibility, and what risks come with using AI in that context. This was done through a descriptive case study, descriptive case studies are suited to answer questions that begin with “why” or “how” (Blomkvist & Hallin, 2014, p. 61). A typical case study focuses on a thing or phenomenon, which could be anything from an organization, to an employee at the organization. The goal with a case study is to gain detailed insight into the study subject and its complexity. This thesis uses a single case to try and answer the research question. A descriptive case study leads to a detailed analysis of a particular phenomenon and its context (Oates, 2025, p. 150).

Critiques against case studies often mention that case studies are primitive and subjective (Blomkvist & Hallin, 2014, p. 61). The case is however generalizable to similar cases where AI or other disruptive technology is being considered as a “force multiplier”. This argument may not hold true for unique case studies, but then the case study is instead interesting and has value simply because the case is unique (Blomkvist & Hallin, 2014, p. 62).

Initially, the aim of this thesis was to conduct a technical sensor analysis of specific ground vehicles within the Swedish Armed Forces. Since the military sector is characterized by strict restrictions regarding transparency and access to sensitive technical data, organizational constraints arose regarding data availability. For this reason, the study was reformulated to adopt a broader systems science and qualitative perspective, which in turn allowed for an investigation of the underlying socio-technical and logistical processes at a more general level.

4.2 Data collection

Data was collected through mainly two interview methods. The first two interviews conducted were unstructured, to be able to explore the theme of AI and its implementation with ILS processes, and the rest were semi structured interviews. The first interviews were however thematically aligned with the theory, even though no real questions were prepared beforehand.

The unstructured interviews laid the groundwork for the interview guide used in the semi-structured interviews. Semi structured and unstructured interviews allow for discovery in the research area, rather than confirming, or checking one’s own hypothesis (Oates, 2025, p. 198). This way of constructing the interviews process is also motivated by Blomkvist & Hallin (2014, p.70), as a way of gaining insight into the research area unconditionally, which may also have contributed in preventing bias from the author.

The interviews were recorded for ease of analysis, however according to Oates (2025, p. 200-201), interviewees may become nervous when being recorded, which could result in them being more restrictive in their answering. The interviewer may have changed as well, becoming more formal, and less conversational. All interviews but the first one was recorded, and all respondents agreed to be recorded when asked. Notes were also taken during all of the interviews; to mark down keywords and the authors own immediate thoughts about the respondent's statements. The first interview may not be fully accurate, but the key statements were noted down, and used to further understand the research issue. As the interviews went on, the line of questioning became more precise, and in turn the answers more relevant for the study.

The interviews were carried out in a multitude of contexts. Most were online interviews, some were conducted at FMVs offices in Stockholm, and some were conducted at the Swedish Defense University in Stockholm. All of the interviews were conducted on the respondent's terms, meaning they could decide time and place.

4.2.1 Data collection reflection

The interviews could be critiqued for being misleading, as the focus is on the respondent's thoughts and opinions about something, which may not correspond to what that something is in reality (Oates, 2025, p. 209). To try and mitigate this, the answers are compared to the existing literature on the topics.

4.3 Data analysis

The interviews were studied thematically through the theoretical framework. Thematic analysis is a method for identifying and reporting themes within data (Braun & Clarke, 2006, p.4). It can highlight similarities and differences across the data set, and sometimes unanticipated insights. It is important to acknowledge that these insights might be preconceived themes that exist in our heads, understanding this preconception may help in achieving a more objective discussion.

Most importantly, thematic analysis can be a useful tool in policy development. As this work in a way aims to support the FMV in their work to develop policy regarding AI use in ILS processes for ground vehicles, it fits well within the stated goal.

4.4 Choice of respondents

The choice of respondent fell on the accessibility the author had to respondents, called convenience sampling. Most of the respondents were accessible due to the authors place of employment during the thesis process, while some were accessible through the authors university. Other respondents, such as those from the defense industry, were first met face to face at expos, and later contacted. This resulted in respondents from academia, the defense industry, the SwAF, and FMV. The author hopes this diversity in respondents provides a decent insight into what some people from these organizations think about the research question.

4.5 Validity, Reliability and Generalizability

To ensure that the research meets high quality standards, this section addresses the reliability, validity, and generalizability of the study. These criteria are essential for assessing the credibility and usefulness of the research.

4.5.1 Validity

This study uses anonymous interviews as a basis for its empirical evidence. This may hurt the studies validity, as their answers cannot be confirmed. Their titles are however listed, which should provide some weight to their statements.

The theoretical framework, military utility, is a well-established framework at the Swedish Defense University, and is used and described in several different courses. By basing the thematic analysis around that theory, it gains validity.

The questionnaire for the semi structured interviews was prepared after two unstructured interviews were conducted. This questionnaire was then assessed by the author's advisor at FMV.

Finally, the thesis was reviewed by the thesis supervisor, and colleagues at the Swedish Defense Materiel Administration, some who themselves hold masters' degrees, and others who have previously been supervisors at the master's level. This expert analysis should strengthen the thesis validity further.

4.5.2 Reliability

Interview studies are inherently unreliable, consistency is hard to achieve across interviews, and respondents may respond differently to different interviewers. Similarly, different respondents in the same organizations may have conflicting views or answers to the ones presented in this study. The findings might therefore not be easily replicable.

In the respondents list there is a distinct lack of AI or machine learning expertise. While most respondents are experts in their respective fields, such as human machine interaction, or ground vehicle capabilities. There were genuine attempts at getting an interview with an AI researcher, however, schedules rarely lined up or meetings were canceled. This may affect the reliability of the study, and the findings may have been different had they been interviewed. To attempt to mitigate this, the discussion on AI use cases is more strictly compared to previous research than other parts of the discussion.

4.5.3 Generalizability

This case itself is rather unique, the respondent's answers are however, found in other studies and supported in the literature. This study may be generalizable from a Swedish context; however, different Armed Forces and their employees may answer differently. Cultural differences or different conditions in regards to how likely decisionmakers are to allow new technology into testing.

4.6 Ethical considerations

As this study is analyzing the use of AI in a military setting, it is important to carefully consider what risks come with AI implementation. After all, the goal with such a system is to increase its availability, and in turn increase its effectiveness, which is ultimately to dispatch enemy forces with lethal methods. Using AI for such a purpose may be controversial.

To minimize risk and to make sure the data is being used responsibly, the interview respondents were informed how the data would be used, and that the participation is completely voluntary, and they have the right to withdraw consent at any time.

The respondents were also given clear information about the purpose of the study, and what potential risks or opportunities are included with their participation. This transparency is crucial to be able to build trust between the interviewer and the respondent, and to make sure the respondents feel secure and comfortable with sharing their knowledge and opinions (Vetenskapsrådet, 2025).

There are inherent risks with digitalization, implementing more electronic devices that collect information may send out unwanted signatures. A risk itself is the digitalizing and the compilation of sensitive data. The data itself could highlight logistical issues for the SwAF, that vulnerability could be exploited by the enemies of the state.

5 Results

This chapter compiles the interviews that were conducted for this study. The first two interviews were done to help formulate the interview guide and to unconditionally research and explore the problem. The goal with the interviews that were conducted was to identify where AI could be used within ILS processes to potentially increase availability. All respondents were informed beforehand of the study's aim, and their rights to withdraw their consent.

The texts below are from academia, the defense industry, the Swedish Armed Forces [SwAF] and the Swedish defense materiel administration [FMV]. In total seven interviews were conducted, the people interviewed could generally be regarded as experts within their field. Among them, the chief scientist of the Swedish Air staff, a Technical Systems Manager for ground vehicles, FMV employees with 20+ years' experience within the organization, where several of the respondents hold doctorate titles. To represent the industrial perspective, a global sales manager from a Swedish small/medium-sized enterprise was interviewed, to gain both the respondents and the organizations point of view on the subject.

These statements have been thematically categorized according to the theoretical framework. Some statements may not be defined in the theoretical framework, but have been interpreted as affecting the research problem, and have therefore been included.

As the interviews were conducted in Swedish the statements below should not be taken as exact quotes by the respondents.

5.1 Empirics /Interviews

There were 7 interviews in total. The respondents are referred to as R1 through R7 to be able to refer to them in the text.

Table 1 Background of interview subject

Interviewee	**Date of interview**	**Background**
R1	2025-03-05	Military and Engineering
R2	2025-03-17	Military logistics and Academia
R3	2025-04-13	Military and Army vehicle acquisition
R4	2025-04-14	Military and Marine acquisition
R5	2025-04-20	Military and Academia
R6	2025-04-22	Defense Industry
R7	2025-04-24	Military ground vehicle system manager

5.2 Effectiveness

Military effectiveness addresses how well AI or machine learning [ML] can improve the operational availability of ground vehicles and support the overall mission.

5.2.1 Temporal aspect

A prominent theme among all respondents is that the most immediate and tangible utility of AI within ILS lies in saving time. Respondent 4 mentions that "*when new systems are introduced, that system comes with new documentation, that documentation is often spread out in different places and software programs"*. Searching for specific maintenance instructions in extensive technical manuals can take hours for maintenance personnel(R7). Implementing an AI-driven search function or database could allow a mechanic to instantly retrieve the correct procedure (e.g., changing a shock absorber), drastically reducing maintenance downtime and thereby increasing availability.

Other respondents from FMV and the SwAF mention that AI could be used as decision support to lower the cognitive load in a stressful situation, to increase the operator's effectiveness. One mentions that it could be used to analyze sensor data, or drone footage to warn human operators when something suspicious pops up (R1). On the other hand, R5 argues that operators risk becoming reliant on the technology, constantly delegating one's own thinking could impair learning and judgment, especially when the technology isn't available.

5.2.2 Automation Bias

While the technical utility of AI is recognized, the respondents raised severe socio-technical risks associated with AI adoption in military systems. One respondent argues that if an AI system constantly assists or makes decisions for human operators, the operators risk losing their fundamental understanding of the task (R5). The same respondent exemplifies this by saying "*an old Volvo 240 is fully mechanical, the laws of nature are more prominent, and the limits are much clearer compared to a factory new Audi RS6 with all wheel drive and traction control. These systems obscure what is actually possible or safe".* Insinuating that there is a distinct danger of operators treating AI outputs as absolute truth, leading to a loss of critical thinking. Respondents emphasized that an AI should function solely as a decision support tool (an advisor), not a decision-maker (R3; R4; R5). R7 mentions that if newly examined engineers or mechanics use AI to bypass traditional learning within ILS, they may develop a blind trust in the system, leading to systemic incompetence over time, saying quote "*An engineer who's never served or doesn't understand how the organization works at the lowest levels, may never be forced to actually learn how it works*".

On the other hand, R6 indicates that systems like this shouldn't impact the human operator at all, and that the cognitive load should preferably be zero. They liken it to telephone operators in the past, manually connecting phone calls using wires, systems like that should perhaps be automated. Arguing that such automation doesn't necessarily negatively affect a human operator's skillset in a negative way, but instead automates tasks where data flows, and where a computer program is more accurate than a human ever will be(R6).

5.2.3 Maintenance and Sensor data analysis

AI Models can be and should be used to analyze maintenance and sensor data (R2; R3; R4; R6; R7). R3 mentioned that sensors such as microphones, accelerometers or thermometers, would provide good data, as they can detect small changes way better than humans, but that a collection and analysis of this data may be regarded as overkill.

Some respondents mention that while it may be useful, for optimal performance this requires the sensor or data collection system to be built in(R4), as ease of retrofit may differ from system to system. Industry respondents say there is hardware available that is "*basically plug and play*", provided that it's been configured for the system beforehand(R6).

Respondents from the Armed Forces and FMV meanwhile highlighted the desire to acquire a system that uses sensors and AI to detect anomalies (R3; R7). Perhaps indicating a lack of cooperation between parties, or decisionmakers simply not knowing the system.

However, this also requires a structured system for data collection, which currently does not exist on a broad scale in the SwAF with the exception of the Air Force according to R3. R5 says that the reason Air force can have predictive maintenance is partly due to their history of recording flight hours, and partly due to the Air domain being quite predictable, "*the air domain usually contains nothing but air, and maybe some birds*." The ground domain is the opposite, there are people, animals, and a bunch of environmental hazards like landslides or rockfalls that affect accessibility, making it harder to know what went wrong and why it went wrong (R5).

5.2.4 AI's Multipurpose

R6 argues that just one AI machine learning model can have a wide range of uses. They describe how their company's model, which was initially built for anomaly detection in IT-applications, found new and different use cases when pitched to different domain experts. Their model could detect anomalies related to component failure in public transport vehicles, decreasing their downtime, but also monitoring of fiber-optic cables for suspicious activity.

5.2.5 Wartime vs. Peacetime Data

A few respondents mention that A major barrier to AI's effectiveness in this context is the quality and relevance of current data (R6; R7). Maintenance data logged during peacetime (e.g., minor collisions during training) does not reflect the failure rates or the specific needs of an actual wartime scenario. To be truly effective, an AI system used in ILS needs to "speak war", according to R7. It must be able to simulate scenarios involving combat losses, high-intensity warfare, and actual wartime consumption to accurately forecast the need for spare parts.

5.2.6 Unmanned Systems and Local Manufacturing

R2 explains that a clear trend from Ukraine is the extensive use of AI Unmanned Ground Vehicles (UGVs) and drones for frontline logistics to reduce the risk of human casualties. Furthermore, the ability to manufacture spare parts locally, such as using micro-factories or 3D printing near the frontlines, is seen as a "game changer" by R2 to maintain vehicle availability

when traditional supply chains are severed. These combined create a more resilient and flexible supply chain. The challenge with this technology is implementing it, and how to integrate them with existing systems, furthermore UGVs need protection against electronic warfare to be fully useful (R1; R2).

5.3 Suitability

Suitability explores how well AI integrates with the existing DOTMPLFI structure within the Armed Forces.

5.3.1 Organizational Culture and Trust

There is cultural friction regarding the adoption of AI and new technology according to some respondents (R1; R4; R7). Historically, there is an over-reliance on human operators, driven by a belief that a human looking directly at reality makes better, unfiltered judgments than a machine (R4). This skepticism often hinders the implementation of systems that could otherwise filter out noise and present critical data more clearly. According to R1 "*one of the biggest risks with AI is the lack of transparency, that it is inherently a black box technology*. This is "scary" to military decisionmakers, as they don't understand it (R1; R4). Presenting the capability or utility of the technology could help in convincing these decisionmakers (R7).

According to R1, a major hurdle for suitability lies in organizational and integration issues rather than the technology itself. Historically, the SwAF has sometimes developed capabilities that ultimately resulted in impaired operational capability (R1). This occurred because commanding officers created integration bottlenecks down to the unit level. Putting "spokes in the wheel" to fight for their own specific solutions rather than aiming for functional synergy wanting "*1+1 to equal 5*" (R1). Although the situation has improved, the sentiment remains. For instance, the SwAF currently utilizes the NATO Architecture Framework (NAF), but according to the R1, it is often applied incorrectly, which further complicates the integration of new systems like AI.

5.3.2 Competence

As Sweden proceeded with the disarmament of the defense, the competence that existed within the organization moved on to other industries (R3). Some respondents liken this to the Swedish defense shooting themselves in the foot (R1; R7), and that it was one of the dumbest decisions in Swedish history (R5). They also indicate that this competence is currently being rebuilt (R2).

5.3.3 Collaboration

The importance of collaboration is mentioned by several respondents (R1; R2; R6). Between the industry, the armed forces and FMV, often called triple helix. R2 mentions that there is a lot of collaboration in regards to logistical solutions, such as unmanned ground vehicles for last mile delivery of supplies. While R1 respondent mentions that there needs to be more collaboration, not only for logistics but in regards to AI implementation in military management systems.

5.3.4 Organizations Do What is Measured

A systemic organizational issue hindering suitability is that "*organizations do what is measured*"(R6). Currently, the SwAF and FMV primarily measure financial costs, but rarely measure the actual operational availability or the alternative costs of a human performing manual tasks (R5; R6). R6 says that without clear performance indicators for availability, it is difficult to motivate the integration and investment in new AI logistics systems.

5.3.5 Skepticism Toward Rapid Change

The rapid pace of technological advancement is itself a source of friction. R1 notes that stakeholders naturally become skeptical when change happens too fast, such as target development times dropping from three days to three minutes in Ukraine. However, regarding the technical risk of AI implementation, R1 was less concerned: "*I don't see it being that big of a risk... If the system is secure, the AI model can roam free in the system."*

5.3.6 Information and Data management

R3 says that the SwAF generally demands that ground vehicles and their maintenance systems remain disconnected from the internet to avoid revealing their positions or creating cyber vulnerabilities Therefore, any AI integration must function within a closed, secure loop. While this creates immense tactical advantages, it also brings its own set of integration challenges according to R1. Looking at the current situation in Ukraine, data processing and AI implementation rely heavily on interlocked systems and continuous information flows.

However, even within a closed system, data standardization is an issue. While some projects utilize structured technical documentation suitable for AI training, R4 noted that military branches often default to ordering unstructured “flat PDFs”. A unified format for receiving technical information would be needed to be able to train a reliable AI model (R4).

5.3.7 Implementation Through Testing

A lesson reinforced by both the Ukrainian context and some respondents (R1; R3; R4; R7) is the approach to implementing new technology: the best way to understand it is to simply start using it. To innovate effectively, forces must test the product or service in the field to discover its actual utility, which units should own it, and at what command level it is most effective (R1). At the same time, it’s important to be somewhat restrictive with implementing new technology so that it doesn’t spiral out of control (R3; R4). R4 stresses that you need to know why you are testing something, and for what reason. Similarly, you need to understand what shortcuts are being taken.

Other respondents mention not having enough time to pursue these endeavors, and that other projects take priority (R3; R7). The innovation is mostly delegated to the industry, and because of the hierarchical culture in the armed forces, conscripts or lower-level officers’ opinions may be dismissed (R1; R4). R1 suggests to change one’s line of work due to this hierarchy, and move to a place where one’s ideas are heard. While R7 argues that there should be people within military organizations responsible for gathering ideas from employees.

5.3.8 Dynamic Logistics vs. Static Systems:

A few respondents with insight into the Ukraine conflict mention that static logistics systems, such as fixed depots and scheduled transport routes, have proven highly vulnerable to modern threats like drones and long-range missiles (R1; R2). To counter this, R2 says logistics must become dynamic, utilizing dispersed routes and high-speed deliveries, where AI can assist in optimizing chaotic supply routes in real-time.

5.4 Affordability

Affordability relates to the life-cycle costs and the resource allocation required to implement AI in ILS processes.

5.4.1 Supply chains and “just-in-time”

The modern supply chain strategy of "just-in-time" (JIT) is heavily criticized by R7 as a peacetime concept that fails in a crisis. Relying on JIT without knowing the actual lead times during a conflict essentially turns it into "no time" says R7.

R7 continues by saying that AI could provide utility by running robust simulations that visually demonstrate to budget decision-makers how much material is actually needed. According to them, military economists often hesitate to allocate budgets for "what-if" scenarios, preferring to avoid taking financial height for unconfirmed risks. Saying that “*AI could potentially help convince budget decision-makers that higher initial investments and spare parts are strictly necessary in wartime”*.

5.4.2 Alternative costs

Military organizations often lack the ability to evaluate things other than money as a cost according to R4. Because conscripts and employees are already paid for, the benefit of automating some of their tasks is hard to measure and understand. Therefore, the true cost of not implementing smarter maintenance systems is often hidden.

A critical breakdown that could have been prevented costs enormous amounts of money (R3). R4 says that it is often viewed as an isolated incident, and the overarching financial loss is not fully recognized by the organization. Furthermore, R4 argues that the operational cost of a vehicle breaking down in the terrain and requiring recovery has severe tactical consequences that extend beyond mere financial metrics.

5.4.3 Cost of sensor analysis

According to R2, research into sensor analysis and how it could be used to improve availability was already being done even in the 90s. They however concluded that it was too expensive to retrofit vehicle systems with sensors. During this time, the Armed Forces was way ahead of the automotive industry in regards to technology, now it’s the other way around (R2).

6 Discussion

This chapter analyzes the results of the qualitative interviews through the lens of the Military Utility theory. The findings are contextualized against previous research to answer how AI and machine learning can affect the availability of ground vehicles within the Swedish Armed Forces.

6.1 Use cases and effectiveness

AI is perceived to be highly effective when applied to narrow, predictable tasks, such as analyzing maintenance or sensor data. Conversely, it is deemed ineffective or risky when applied to strategic decisions or highly unpredictable environments. Mojtahed et al. (2024, p.31) argues that AI data processing should be used where there are consistent flows of large amounts of data, to maximize its utility within the SwAF, respondents generally agree with this sentiment.

Some of these recommendations are from the user perspective, which may skew the discussion to favor their point of view. Steve Jobs famously said "*A lot of times, people don't know what they want until you show it to them*." Their experience of the problem is more interesting to investigate and solve, rather than taking what they say at face value and running with it.

6.1.1 Time Savings and Searchable Documents

The empirical evidence demonstrates that the most straightforward utility of AI within ILS processes lies in the temporal aspect of maintenance. As some respondents noted, a significant amount of time is currently spent searching for specific procedures across extensive, fragmented technical manuals and documentation. Implementing an AI-driven search function or database using natural language processing could allow a mechanic to instantly retrieve correct repair procedures, significantly reducing MTTR. This capability already exists in generative tools like NotebookLM, which can summarize and make uploaded text documents searchable with natural language. This aligns with several studies, specifically Alenljung et al. (2026), where AI was used in a military planning exercise to be able to search through documentation and summarize it. This use case also aligns with Schraagen (2024), who argues that AI should be utilized to automate dull, dirty, and dangerous tasks, thereby freeing up scarce personnel to focus on more cognitively demanding operations.

6.1.2 Predictive Maintenance and Sensor Analysis

Predictive maintenance (PdM) through machine learning and sensor data analysis featured prominently as a primary desire for increasing ground vehicle availability. Respondents highlighted the potential of sensors to detect small physical anomalies before critical breakdowns occur, such as microphones or accelerometers. Parameters like vibrations are highly accurate indicators of poor health, which is confirmed by Molęda et al. (2023). These types of sensors can more accurately detect small differences in their respective measurement. Newer systems have sensors built in, retrofitting sensors to old systems needs to be done at a reasonable and measured cost.

The Air domain has had predictive maintenance for some time C3 AI. (2025), but successful predictive maintenance implementation does exist in military domains as well. For instance, the Spanish Armada's SOPRENE project (Fernández-Barrero et al., 2021), where PDM was done on Frigate engine bays. Or Esteban et al. (2026, p.22) study on how PDM affects military trucks. Both of these relied on technical solutions that won't win any favor with the SwAF. One relying on transport of sensor data

On the other hand, an anomaly detection system is functionally useless if the surrounding logistics infrastructure is broken. Which was exemplified by the US Army's $135 million proof of concept for the Stryker APC fleet, built to tailor maintenance schedules. Despite this technical capability, the GAO (2025) found that most US Army vehicles failed to meet combat readiness requirements due to an overall decline in actual maintenance, severe supplier shortages, single-manufacturer constraints, and a lack of technicians well-versed in physical drawings. Their problems essentially encompass the whole ILS structure, attacking the problem from a more holistic perspective may have helped.

Thus, vehicle availability depends on the capacity to execute physical repairs, not just digital predictions. Additionally, AI data processing depends heavily on high data quality, completeness, and validity. Peacetime training data does not reflect the failure rates or high-intensity consumption of actual warfare.

6.1.3 Dynamic Logistics

Lessons drawn from the war in Ukraine underscore that effective combat logistics play a greater role in achieving tactical victory than developed strategies, as logistics serve as the backbone of military power (Ti & Kinsey 2023). Static logistics systems, such as fixed depots and scheduled transport routes, have proven highly vulnerable to drones and long-range missiles. Consequently, logistics must become dynamic, utilizing dispersed routes and high-speed deliveries. AI can optimize chaotic supply routes in real-time to facilitate resilient front-line supply chains, supported by autonomous trends like UGVs replacing humans in dangerous environments (Livingstone, 2026) and local 3D printing near the frontlines to remove soldiers from danger.

6.1.4 AI multi-use

The potential of AI machine learning to be a multi-purpose tool is highlighted in the results chapter. The defense industry explains how a single anomaly detection model can detect anomalies in all sorts of systems. This could prove more attractive to military decision makers, as it might seem more straightforward to acquire one multipurpose system, rather than several systems for each identified use case.

It’s also more useful to have one system, addressing the concern from one respondent regarding the new system having more interfaces and documentation in different places. On the other hand, the use still relies on decisionmakers being open to explore what sort of capabilities or use cases there are for such an AI model.

6.2 Suitability for the SwAF

According to the Military Utility framework, an element of interest only possesses true utility if it achieves a high degree of suitability within the military actor's existing DOTMPLFI structure. The empirical data reveals that the primary friction surrounding AI integration is organizational and cultural rather than strictly technical. Military decision-makers express deep skepticism toward AI because it is inherently a "black box" technology lacking transparency. This systemic resistance is compounded by the low technical organizational maturity of the SwAF. According to Mojtahed et al. (2024) the SwAF lacks the required internal competence and exhibits reluctance to implement new technologies. That paper is a few years old, and conditions may have changed within the SwAF, but the sentiment is still echoed by Ellmén (2024), and some respondents.

The respondents paint a somewhat bleak picture of SwAF, some blaming the eternal peace. On the other hand, the SwAF shows signs of wanting to change. A stated goal for the SwAF is to be more data driven (Mojtahed et al., 2024). There are tests regarding UGV's, (FMV, 2025), and several initiatives to build technical competence, such as a cyber defense division, and investments are pouring in from the Swedish Government. Defense Authorities are also some of the most attractive employers in Sweden currently (Universum, 2025), and that the future is looking brighter and brighter. The potential future problem will be to keep that competence within defense organizations, and to keep them from leaving to the private sector.

6.2.1 Cooperation and organization.

The empirical evidence suggests that, for an effective incorporation of artificial intelligence, cooperation between the industry and the armed forces is needed for the specific purpose of AI implementation. Shumlianskyi et al. (2025, p.11) also argues that training and cooperation with industry is of necessity when incorporating AI. How well cooperation works in the triple helix is split between the respondents, some respondents mentioned that there was sufficient cooperation between the industry and the defense authorities, while others argued that more cooperation was needed to be able to implement AI and new technologies.

The importance of Cooperation is also highlighted by other studies. Jonsson et al. (2023, p.49) mention that a key issue in cooperation between authorities is the lack of a common communications system between the defense SwAF. Confidential information may be the reason for this choice, it also makes it harder to share that information between authorities.

While studies or reports from FOI may affect or help the FMV or SwAF understand their situation, the employees simply don't have time to read and address the issues (Jonsson et al. 2023, p.51). Some respondents suggested that a team should have that specific responsibility, while others suggest that the employees should have dedicated time to work on their ideas irrespective of their workload. This setup is common in companies that develop products, often incorporating ideas from the whole organization, from janitors to the CEO. This problem may stem from the organizations being inherently hierarchical. Others suggest it has to do with the "eternal peace" Sweden experienced and the almost total disarmament of the Swedish Defense.

6.3 Affordability

6.3.1 Measurements

An issue presented by the respondents is that "organizations do what is measured". Currently, the SwAF and FMV primarily measure financial procurement costs, while rarely measuring actual operational availability or the alternative costs of a human performing manual tasks. Without clear performance indicators for availability or Total Ownership Cost (TOC), motivating integration and investment in complex AI systems remains exceptionally difficult.

6.3.2 Supply Chain Management

A prominent finding regarding the affordability and resource allocation dimension is the explicit critique of modern "just-in-time" (JIT) supply chain strategies, which respondents classified as an inherently vulnerable peacetime concept that fails during a crisis. Relying on JIT without verified lead times during a conflict essentially means "no time". Here, AI could provide strategic utility by running robust simulations that visually demonstrate material shortfalls to budget decision-makers, who often hesitate to fund unconfirmed "what-if" risks. This could allow AI to help convince economists that higher initial investments and stockpiles are strictly necessary for wartime capability, similar to how the US Defense Logistics Agency (DLA) currently utilizes AI to predict material demand and monitor supply chain risks.

6.4 Risks

Some of the biggest issues found in previous research, is that in AI data processing, the data part is the most important. The gathered data needs to be of high quality, meaning of high accuracy, completeness, consistency, and validity, the maintenance data cannot be vague. Similarly to jetfighters, perhaps flight hours, and mission type is necessary for the data to achieve high quality.

This data then needs to be securely stored. However, this should not be an issue of data security, as the armed forces most likely already know great ways to manage and store data. If the model is compromised then it's already too late.

But there are also risks with not compiling our data. It creates a lack of useful, researchable data, which in turn hampers the development of technology and tools that could be of use to the SwAF.

In Ukraine however, the number of cyber-incidents is decreasing, and no critical cyber incidents were recorded in the second half of 2025 (Ukrainian National News 2025). Suggesting that in the start of a conflict, digital infrastructure may be more vulnerable, but if the conflict is drawn out, the cycle of measure and countermeasure starts rolling, and the advantage gained may be lost.

6.4.1 Automation Bias

However, the results also highlight severe socio-technical risks associated with automated decision support, primarily concerning the phenomenon of automation bias. This bias comes from over-relying on automated systems, rather than critically assessing its output or task. Non-technical risks with AI, which could be argued are less obvious and potentially more dangerous than technical risk (Shumlianskyi et al. 2025, p.1).

One respondent illustrated this danger by comparing a fully mechanical Volvo 240, where the laws of nature and physical limits are highly prominent, with a modern Audi RS6 equipped with multiple driver aids, which obscure what is actually safe or possible. If newly examined engineers and mechanics utilize AI to bypass traditional learning within ILS, they risk developing a blind trust in the system, leading to systemic incompetence over time. Wickenberg-Bolin et al. (2025) emphasize this critical vulnerability, showing that under pressure, AI exhibits a clear bias to favor initial algorithmic outputs rather than evaluating situations critically. Therefore, to prevent a disastrous decline in practical understanding, AI must function strictly as an advisory decision-support tool rather than an autonomous decision-maker.

The challenge is teaching the operator to use the technology but not to rely on it. Advanced diagnostic systems work well in fully equipped civilian workshops, but a soldier "knee-deep in mud" during combat needs practical understanding and improvisation skills. Simkute et al. (2024) argues that LLMs makes easier tasks easy, and harder tasks harder, as users have to learn how the system works but also how the automation works, putting a bigger cognitive strain on the operators.

6.5 Critical reflection

This area of research is naturally confidential, limiting the amount of research possible and limiting the potential insight gained, several of the interviewees mentioned that they cannot disclose information regarding current “operative” AI machine learning use. Perhaps the armed forces have sufficient or even great AI competence within the organization, but that is not disclosed.

This thesis was done at the request of the Swedish defense materiel acquisition agency [FMV], it’s possible this thesis is biased in their favor, and that the opinions of the people employed at the agency are reflected in this thesis, instead of being a more objective piece. While employees at FMV were interviewed due to ease of access, respondents from academia, the defense industry, and the SwAF were included to provide a balanced overview.

There is also a possibility that the industry tries to hype up their own technology, to appear more useful or have more use cases than in actuality. Even though the author is just a student, there is a clear affiliation with FMV, which may skew their answers.

Quite a few of the cited works come from the Swedish Defense Research Agency [FOI], and all of the interviewees were Swedish speaking, Swedish citizens. Which may provide a decent Swedish point of view on the issue, but makes it less generalizable.

As this thesis was being written, several situations impacting the thesis occurred. Sweden introduced an AI policy, the USA and Iran conflict, technological advancements in AI. Furthermore, this thesis is heavily affected by the war in Ukraine, many sources point to experiences drawn from that conflict.

One important thing to keep in mind is that these benefits may only be perceived benefits, even though other practical studies have found results, they cannot be confirmed in this study. However, the findings lie in accordance with other studies, researching similar topics.

6.5.1 Theory

The choice of theory may not fit this sort of study. The author of the theory even mentions that AI is a difficult technology to assess in regards to utility. To mitigate this the use of AI was specific to ILS processes for ground vehicles.

6.5.2 Method

The letter of consent was not used to the extent it could've been. Most of the interviews were online, therefore consent was instead achieved verbally.

During the interviews, the author had expected technical answers. Most of the interviews gave overarching views on the topic, perhaps they did not want to give more specific answers due to secrecy, or due not having the technical knowledge.

7 Conclusion

This chapter provides the conclusions and recommendations suggested by the author based on previous research and the results from the interviews. Please observe that these conclusions are of a general nature.

7.1 Conclusions

The aim of this thesis was to examine how artificial intelligence and machine learning can affect the operational availability of ground vehicles within the Swedish Armed Forces (SwAF) through Integrated Logistics Support (ILS) processes. Based on the empirical findings from seven expert interviews and a thematic analysis using the Military Utility framework, the following research questions were investigated.

How can AI create military utility in ILS processes?

The study concludes that AI possesses significant potential to create military utility within ILS, primarily by optimizing the temporal and logistical aspects of maintenance. However, this utility is highly conditional and depends on how well it can be integrated into the existing military organization. The utility is manifested through the three dimensions of the theoretical framework:

- **Military Effectiveness through Optimized Maintenance:** AI demonstrates a high potential to increase operational vehicle availability by generating time savings. Concretely, this could be achieved by implementing Large Language Models (LLMs) to make extensive technical documentation instantly searchable, drastically reducing the Mean Time to Repair (MTTR). Furthermore, utilizing AI for Predictive Maintenance via sensor data analysis allows for the early detection of anomalies, shifting maintenance from reactive to proactive before critical breakdowns occur. UGVs show promise as a logistical solution for last mile delivery in dangerous areas.

- **Affordability, Dynamic Logistics and Simulations:** AI creates financial and operational utility by automating resource allocation. By running robust simulations for material demand, AI could identify vulnerabilities of peace-time supply chains, and be optimized for war time logistics. In practice, AI could optimize chaotic and dispersed supply routes, creating dynamic Logistics, when traditional supply lines are severed.

- **Military Suitability and Organizational Friction:** Despite high potential effectiveness, the actual military utility is currently constrained by a low degree of Military Suitability within the SwAF's DOTMPLFI structure. Technically, strict information security mandates that maintenance data systems remain in a closed loop, disconnected from the internet. Organizationally, suitability may be restricted by poor data standardization for ground systems, and a cultural skepticism toward "black box" technologies among military decision-makers. While there is a stated willingness to become more data-driven, bridging the gap between technical capability and organizational trust remains the largest hurdle for creating true utility.

What are the risks with becoming more data-driven?

While the utility is evident, becoming more data-driven and implementing AI introduces severe technical, operational, and socio-technical risks:

- **Data Quality and the "Peacetime Bias":** A major operational risk is that current maintenance data is logged during peacetime training and does not reflect wartime failure rates or high-intensity consumption. For AI to be effective, it must "speak war"; otherwise, models will generate inaccurate predictions in a conflict scenario.

- **Information Security Vulnerabilities:** Compiling and centralizing sensitive logistical and vehicle health data creates a high-value target. If compromised, this data could expose systemic operational vulnerabilities to adversaries.

- **The Ironies of Automation and Competence Loss:** The most prominent socio-technical risk is the phenomenon of automation bias, where users develop a blind, over-reliance on digital tools. If AI constantly drives decision-making, personnel, especially newly trained engineers and mechanics, risk losing their fundamental mechanical understanding and critical thinking skills. This is a critical vulnerability in combat environments where digitalization might fail, and soldiers must be able to rely on practical improvisation.

Conclusively, to leverage the military utility of AI safely, the SwAF should treat these systems strictly as advisory decision-support tools rather than autonomous decision-makers, ensuring the human remains firmly in the loop.

7.2 Further Research

Further research into the practical applications of AI within military logistics is essential to confirm and expand upon the findings of this study. Specifically, a practical study exploring how anomaly detection algorithms work in the field would be of high interest. Such research could evaluate whether these systems are as dynamic and adaptable in real-world environments as the present results suggest.

A deeper dive into the risks with automation in the military field to either confirm or deny the suggestions made by the respondents could be of use when evaluating what to automate.

Future studies should compare the effectiveness and risks of using autonomous platforms, such as Unmanned Ground Vehicles (UGVs), against conventional logistics methods, particularly considering the need for dynamic supply chains and the identified vulnerabilities of static supply routes in modern conflicts.

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9 Appendix

9.1 Appendix 1. Interview guide

A short background about the thesis, and the author. The thesis investigates how AI can be used within ILS to increase their availability. Definition of Key concepts (AI and ILS) are defined by the author and the respondent.

1. Background & Current Role

What is your current role and background?

What are you investigating or working on currently?

Are you analyzing this data to find patterns or improve availability?

What are the main challenges with vehicle availability and the ILS processes today?

3. Future Technologies & AI Implementation

How do you see AI and machine learning being used within ILS to increase availability?

4. Risks, Human Role & Automation Bias

What risks do you see with digitalizing these vehicles and relying more on AI?

5. Culture, Organization & Innovation

In your experience, is there a cultural skepticism towards AI and new technology within the organization? Why do you think that is?

Are we doing enough testing of new technology in Sweden? What are the barriers preventing us from trying new things faster?

6. Lessons from Ukraine & Peacetime vs. Wartime Logistics

What lessons can we learn from the ongoing war in Ukraine regarding logistics, adaptability, and maintenance?

How does the current "just in time" peacetime logistics differ from actual wartime needs for spare parts?

7. **Conclusion**

Is there anything you have thought about that you haven't mentioned, or anything else I should have asked?

Can you recommend anyone else I should interview for this thesis?

Thanks!

9.2 Appendix 2. Letter of consent

Information and consent form for processing personal data associated with student essays

To process personal data, a consent must be obtained that clearly and in a distinct way states what data will be collected and what it will be used for. This information and consent form explains how the personal data will be processed and contains contact details.

The personal data is processed with **your explicit consent**. Participation in the study is completely voluntary. You can withdraw your consent at any time without giving a reason. If you do not consent to the processing of personal data, you can do so without suffering negative consequences.

How will the personal data be used?

This is a study in system science for degense and security, performed as a case study on the Swedish Armed Forces. The purpose of this study is to research the military utility of AI within integrated logistics support.

Interviews are carried out to gain expert knowledge. This interview is expected to take **up to 1 hour**.

The study is carried out within the scope of the course "Master's Thesis, Systems Science for Defense and Security VT2026". The study is an examination task and will be reported as an essay that constitutes an independent scientific work. Results will be presented in consolidated form, with interview quotes as illustrations or clarifications. Data will be transcribed into written language and presented in such way that it will not be traceable to an individual.

Disclosure pf personal data to third parties: This thesis is being written in collaboration with the Swedish Defense Materiel administration, your personal data may be shared with the students supervisor at that agency.

What personal data will be processed?

Your voice will be recorded for ease of transcription, and your name will be stored on this consent form.

How is your personal data protected and stored?

The recording will be stored on a personal smartphone with a biometric password.

When the personal data no longer is needed for the study's purpose, it will be deleted, this means the recorded audio file, the interview transcripts, and the signed consent form.

Your rights

If you need to have incorrect information corrected, supplement with missing information (correction) or if you regret your consent (revocation), you can contact the responsible student and/or their supervisor in the first instance (see contact details below). You can also contact the SDU data protection officer at dataskyddsombud@fhs.se.

Contact details for responsible student(s) and supervisor

System Science for Defense and Security, Defense Systems.

Supervisor: Johan Granholm – Johan.Granholm@fhs.se

Author: Theodor Hellqvist – Theodor.Hellqvist@student.fhs.se

The personal data controllers are Swedish Defence University (SDU), tel. 08-553 425 00. Email registrator@fhs.se

If you want to know more about how SDU as an authority handles personal data

If you are not satisfied with how SDU handles your personal data, you always have the right to submit a complaint to the Swedish Data Protection Authority (IMY) via e-mail imy@imy.se or telephone +46 8-657 61 00.

By signing below, I confirm that I have read the above information and understand how my personal data may be processed. I am aware that my participation is completely voluntary and that I can cancel my participation in the study without giving any reason.

Place and date

Name clarification

Signature